2-step PCR mixture and conditions (Barcoded-head primers for seqs pooling)

Huang MEGA Lab (NCKU)

(Lab plain-PDF edition, rebuilt 2026-10-20)

Source of record: Huang MEGA Lab (NCKU). Originally authored on protocols.io (dx.doi.org/10.17504/protocols. created 2025-09-03, version 0. This PDF is the lab's self-hosted copy and supersedes the protocols.io export.

Abstract

PCR mixture and condition (2X SUPERGREEN PCR MASTER MIX)

Procedure

Procedure

---

1. Wear glove, clean up the working bench w. 1% bleach

For 1' PCR head-primers

---

2. Prepare 1' PCR master mixutre for **head-primers** (**prepare 1.2X of solutions for pipetting error if needed**)

PCR mixture for head-primers for each reaction

Component	Volume	Volume (1.2X)	Final conc.
PowerPol 2X PCR Master Mix	**12.5 µL**	**15 µL**	1X
Forward Primer (10 µM)	**0.5 µL**	**1.2 µL**	**0.2 µM**
Reverse Primer (10 µM)	**0.5 µL**	**1.2 µL**	**0.2 µM**
ddH2O	**10.5 µL**	**11.1 µL**	-
Total volume	**24 µL**	**28.5 µL**	

Component	Volume	Volume (1.2X)	Final conc.
2X KAPA HiFi HotStart ReadyMix	**12.5 µL**	**15 µL**	1X
Forward Primer (10 µM)	**0.75 µL**	**0.9 µL**	**0.3 µM**
Forward Primer (10 µM)	**0.75 µL**	**0.9 µL**	**0.3 µM**
ddH2O	**10 µL**	**12 µL**	-
Total volume	**24 µL**	**28.8 µL**	-

※ **Note:** Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

---

3. Mix the 1' PCR master mixture gently by pippeting. Quick spin the tube.

---

4. Transfer **24 µL** 1' PCR master mixutre in 8-strip PCR tubes.

---

5. Add **1 µL** DNA template in 8-strip PCR tubes, resulting in a **25 µL** reaction mixture for 1' PCR.

※ **Note: Negative control** contains only **24 µL** master mixture but not DNA template

---

6. Mix the reaction mixture gently by tapping the tubes. Quick spin the tubes.

---

7. Carry out PCR using the following condition:

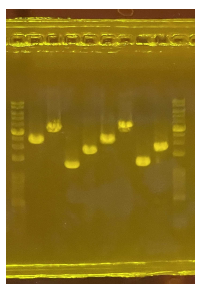
1' PCR condition for **head-primers**

Step	Temp	Sec	Cycle
Initial denaturation	**95 ºC**	180	
Denaturation	**95 ºC**	30	**25 cycles**
Annealing	**60-66 ºC varied (b)**	30	
Extension	**72 ºC**	180	
Final extension	**72 ºC**	420	
Preservation	Preservation	**4 ºC**	∞

b. Annealing varied, 60-66C is working; Refer to 1' PCR primers for annealing temperature c. 1kb ~ 1min extension; enough time allow full extension of sequence

---

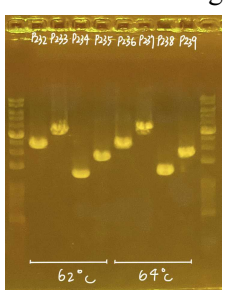
8. Carry out **electrophoresis** for inspection of DNA products



Gel before markdown

---

9. Markdown wells and upload the pictures to the Lab Google drive



Marked gel picture go to the Lab Google drive

For 2' PCR barcoded-head primers

---

10. Prepare 2' PCR master mixutre for **barcoded-primers** (**prepare 1.2X of solutions for pipetting error if needed**)

PCR mixture for barcoded-primers for each reaction **(NO PRIMERs at this point!!)**

Component	Volume	Volume (1.2X)	Final conc.
ZEJU PCR Master Mix	**7.5 µL**	**9 µL**	-
ddH20	**5.55 µL**	**6.66 µL**	-
Total volume	**13.05 µL**	**15.66 µL**	-

※ **Note:** Negative control ALWAYS NEEDED! For example, if you have 5 PCR reactions to run, prepare master mixture for 6 reactions (5 DNA template + 1 negative control).

---

11. Mix the 2' PCR master mixture gently by pippeting. Quick spin the tube.

---

12. Transfer **13.05 µL** of the 2' PCR master mixture to 8-strip PCR tubes.

---

13. Add **1.2 µL** **pre-mixed barcoded-head primers** (Forward + Reverse) to each PCR tubes.

---

14. Add **0.75 µL** of **1' PCR product as template**, resulting in **15 µL** reaction mixture for 2' PCR.

Negative control contains only **14.25 µL** master mixture and premixed barcoded-head primers but not DNA template

---

15. Mix gently by tapping the tubes. Quick spin the tubes.

---

16. Carry out 2' PCR using the following condition:

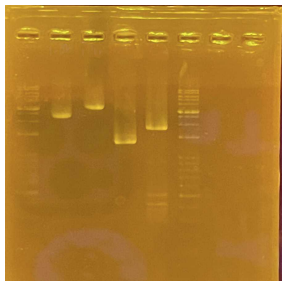
2' PCR condition for **barcoded-head primers**

Step	Temp	Sec	Cycle
Initial denaturation	**98 ºC**	30	
Denaturation	**98 ºC**	15	**12 cycles**
Annealing	**64-68 ºC varied (a)**	15	
Extension	**72 ºC**	20 (b)	
Final extension	**72 ºC**	210	
Preservation	Preservation	**4 ºC**	∞

a. Annealing varied, 65 C is working based on test on 220531; Refer 2' PCR primers for annealing temperature b. 1kb ~ 1min extension; enough time allow full extension of sequence

---

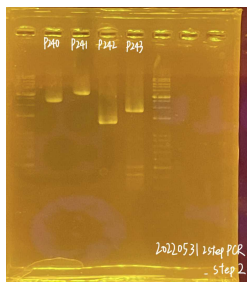
17. Carry out **electrophoresis** for inspection of DNA products



Gel before markdown

---

18. Markdown wells and upload the pictures to the Lab Google drive



Marked gel picture go to the Lab Google drive